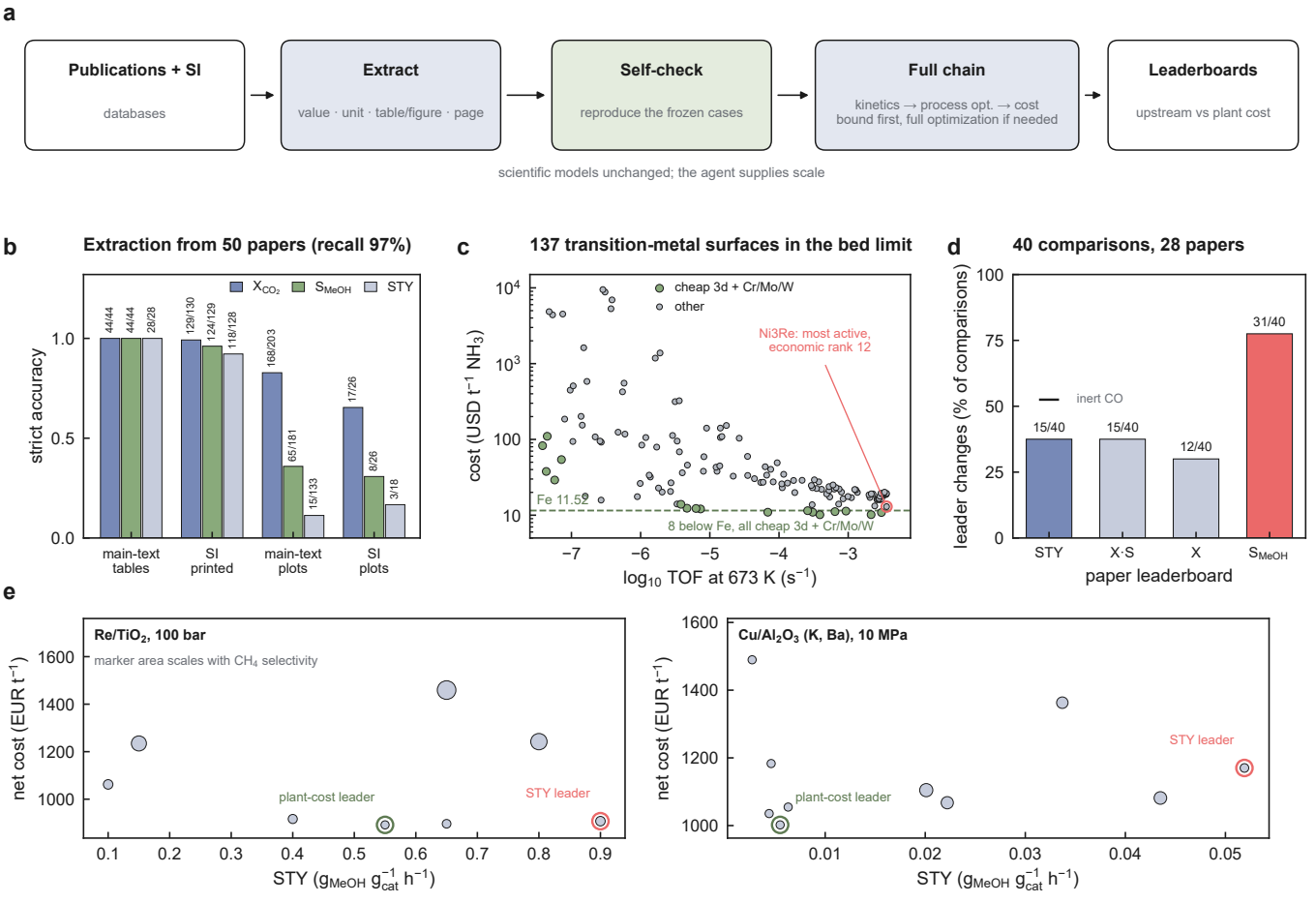

a
Publications + SI
databases
Extract
value · unit · table/figure · page
Self-check
reproduce the frozen cases
Full chain
kinetics → process opt. → cost
bound first, full optimization if needed
Leaderboards
upstream vs plant cost
scientific models unchanged; the agent supplies scale
b
Extraction from 50 papers (recall 97%)
strict accuracy
X_CO2
S_MeOH
STY
44/44
44/44
28/28
129/130
124/129
118/128
168/203
65/181
15/133
17/26
8/26
3/18
main-text tables
SI printed
main-text plots
SI plots
c
137 transition-metal surfaces in the bed limit
cheap 3d + Cr/Mo/W
other
Ni3Re: most active, economic rank 12
Fe 11.52
8 below Fe, all cheap 3d + Cr/Mo/W
cost (USD t⁻¹ NH3)
log10 TOF at 673 K (s⁻¹)
d
40 comparisons, 28 papers
leader changes (% of comparisons)
inert CO
15/40
15/40
12/40
31/40
STY
X·S
X
S_MeOH
paper leaderboard
e
Re/TiO2, 100 bar
marker area scales with CH4 selectivity
plant-cost leader
STY leader
net cost (EUR t⁻¹)
STY (gMeOH gcat⁻¹ h⁻¹)
Cu/Al2O3 (K, Ba), 10 MPa
plant-cost leader
STY leader
net cost (EUR t⁻¹)
STY (gMeOH gcat⁻¹ h⁻¹)